

Discussion 5: Monty Hall Problem

Wednesday, April 29, 2026

Complete this worksheet in a group of **3-5 students**. Write your answers **by hand**, then take a picture and submit it to Canvas. When you submit, only **one person** from your group needs to submit, but make sure to specify who is in the group.

The **Monty Hall problem** is a famous probability puzzle based on a game show. After a contestant picks one of three doors, the host reveals a non-prize from behind one of the remaining doors and asks whether the contestant wants to switch to the other still-closed door. The surprising part is that switching is actually the better strategy. This became especially famous in the 90s when magazine columnist (and person with the highest recorded IQ) Marilyn vos Savant explained the answer in her column and many (mathematically advanced!) readers insisted she was wrong. In this activity, you will test the problem for yourself by simulating it many times and looking at the results.

Materials

You should have three paper cups, and one “prize” (a sticker).

Group roles

Assign each group member a role. You may switch throughout the game if you wish.

- **Host:** The game show host.
- **Contestant:** The player who chooses cups.
- **Recorder:** Writes down the results of each round in the table on the next page.
- If your group has more than 3 people, extra group members can help check that the host follows the rules correctly and help reset the materials each round.

Instructions

1. The **contestant** closes their eyes or turns around while the **host** sets up the cups.
2. The **host** places three cups upside down in a row, hiding the prize under one of the cups. The **host** *must remember* where the prize is.
3. The **contestant** opens their eyes/turns around and selects a cup.
4. The **host** reveals what's under one of the *other* cups, and it must be empty. In other words: if the **contestant** chose the prize cup, the **host** may reveal either of the two other cups, but if the **contestant** chose an empty cup, the **host** must reveal the only other empty cup.
5. The **contestant** is given the choice whether to keep their original cup or to switch cups.
6. The **recorder** notes the **contestant's** decision in the table on the next page.
7. The **host** reveals what is under the **contestant's** final choice of cup.
8. The **recorder** writes down whether the **contestant** got the prize or not.
9. Begin again from step 1, changing roles if you wish.

Results Table

Round #	Did the contestant switch?	Did the contestant win?
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		

Proportion of wins given that the contestant **did not** switch:

Proportion of wins given that the contestant **did** switch:

1. Do you agree that switching seems to be the better tactic based on your “simulation”? Do you think you need more information? Why or why not?

2. In truth, the probability of winning given that one does not switch is $\frac{1}{3}$, and the probability of winning given that one does switch is $\frac{2}{3}$. Let the event that the contestant switches be denoted S , and the event that the contestant wins be denoted W . Write these probabilities in the probability notation we learned in class.

3. Using the probabilities given in (2), can you compute the marginal probability of winning this game, unconditional on whether or not the contestant switches? If yes, compute it; if not, explain what extra information you would need. Be sure to use the probability notation we learned in class.

4. For most people, it is counterintuitive that the probabilities of winning are not 50/50 whether or not the contestant switches. Can you find some intuition for why switching is the better strategy?